Python for Instrumentation and Laboratory Experiments

Why using Python?

- open source
- multiplatform (OSX, Windows, Linux)
- free (no cost)
- used on modern environments (synchrotron for example, but not only)

Advises:

install Anaconda distribution: https://www.anaconda.com/distribution

to list packages already installed:

conda list

to install more packages (for instance pyserial and pyvisa)

conda install pyserial, pyvisa
(or pip3 install…)

VISA : Virtual Instrument Software Architecture

Python for laboratory experiments and instrumentation

Communication computer/instrument

usually need for a physical connection (cable)

modern computer, several ports: USB, thunderbold, ethernet ports
on older computers, RS232

modern equipment: same port as modern computers
on older equipment: RS232, GPIB

of course, often needs for adaptors

Python for laboratory experiments and ir

Communication computer/instrument

of course, often needs for adaptors

https://docs.rs-online.com/c6df/0900766b80da7102.pdf

FTDI (Future Technology Devices International)

driver: https://ftdichip.com/drivers/



Câble USB vers RS232 1m Black with LED

Dans Convertisseurs série et…

Code commande RS 687-7806

Marque FTDI Chip

Référence fabricant **US232R-100-BLK**

20,96 €

la pièce

1

Ajouter

Python for laboratory experiments and instrumentation

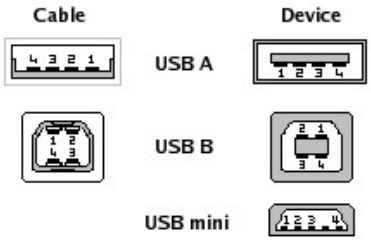
Example: Keithley 2000



Python for laboratory experiments and instrumentation

USB

4 wires



Pin	Signal	Color	Description
1	VCC		+5V
2	D-		Data -
3	D+		Data +
4	GND		Ground



Python for laboratory experiments and instrumentation

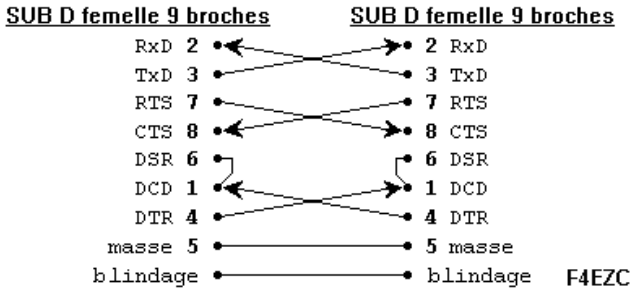
RS232

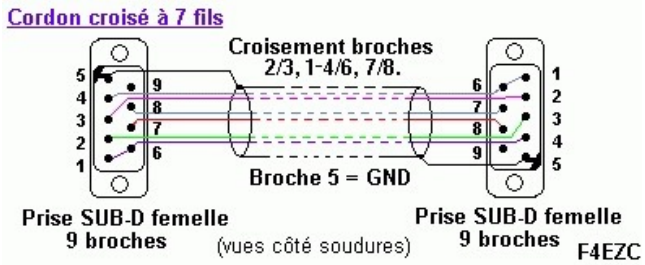
DB9 connector



Voltage:

	Cordon, DB9 femelle vers DB9 femelle, longueur 1.8m, 9 voies Code commande RS 812-926 Référence fabricant 11.01.9018-50 Marque Roline Comparer	5,31 € la pièce 1 Unité Ajouter	DB9 femelle	DB9 femelle	1.8m
	Cordon, Fiche mâle DB9 vers Fiche femelle DB9, longueur 6m Code commande RS 394-3326 Référence fabricant 11.01.6260-25 Marque Roline Comparer	12,23 € la pièce 1 Unité Ajouter	Fiche mâle DB9	Fiche femelle DB9	6m
	Cordon, Fiche mâle DB9 vers Fiche femelle DB9, longueur 1.8m Code commande RS 777-621 Référence fabricant 11.01.6218-50 Marque Roline Comparer	10,08 € la pièce 1 Unité Ajouter	Fiche mâle DB9	Fiche femelle DB9	1.8m





2 and 3 for transmission, 5 for ground

Signal between: -12 to -12 V up to -3 to 3V

More info: https://arcelect.com/rs232.htm

Python for laboratory experiments and instrumentation

GPIB: General Purpose Interface Bus … for parallel communication



Modèle: **GPIB-USB-HS+**

Numéro de référence: 783368-01

Fiche technique et spécifications

~~€ 675,00~~ 1080,00

~~€ 750,00~~

1

VOIR LES ACCESSOIRES

Ajouter au panier

Demander un devis

https://www.ni.com/pdf/manuals/374681a.pdf

drivers: https://www.ni.com/fr-fr/support/downloads/drivers.html

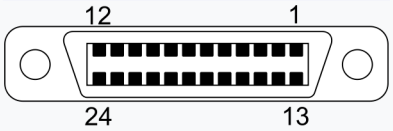




https://stantronic.com/pdffiles/Prologix/PROLOGIX-PL-GPIB-USB-D.pdf drivers: www.ftdichip.com

IEEE 488

Pin out



Female IEEE 488 connector

Pin 1	DIO1	Data input/output bit.
Pin 2	DIO2	Data input/output bit.
Pin 3	DIO3	Data input/output bit.
Pin 4	DIO4	Data input/output bit.
Pin 5	EOI	End-or-identify.
Pin 6	DAV	Data valid.
Pin 7	NRFD	Not ready for data.
Pin 8	NDAC	Not data accepted.
Pin 9	IFC	Interface clear.
Pin 10	SRQ	Service request.
Pin 11	ATN	Attention.
Pin 12	SHIELD	
Pin 13	DIO5	Data input/output bit.
Pin 14	DIO6	Data input/output bit.
Pin 15	DIO7	Data input/output bit.
Pin 16	DIO8	Data input/output bit.
Pin 17	REN	Remote enable.
Pin 18	GND	(wire twisted with DAV)
Pin 19	GND	(wire twisted with NRFD)
Pin 20	GND	(wire twisted with NDAC)
Pin 21	GND	(wire twisted with IFC)
Pin 22	GND	(wire twisted with SRQ)
Pin 23	GND	(wire twisted with ATN)
Pin 24	Logic ground	

Python for laboratory experiments and instrumentation

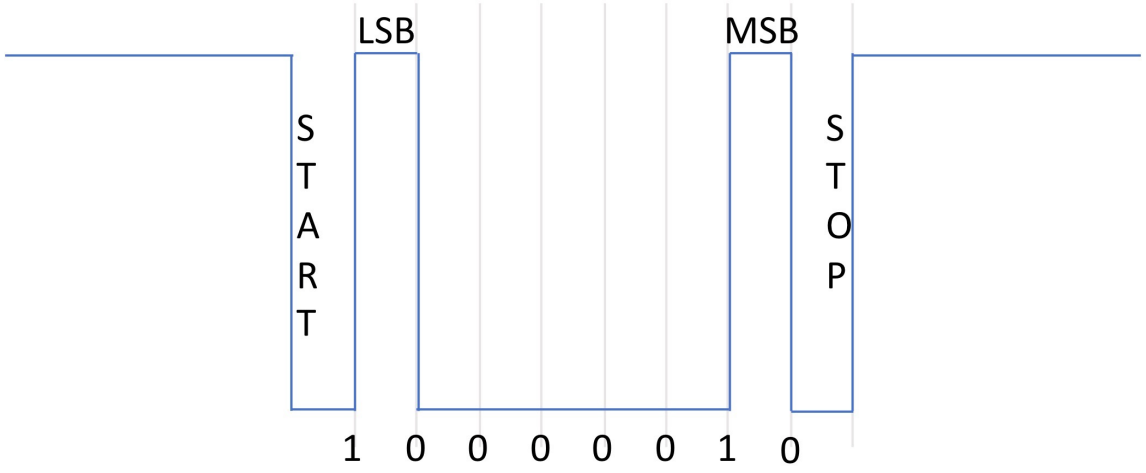
Serial communication

asynchronous

startbit (voltage drop) to start the communication

stopbit, to stop the communication

in between: one byte (8 bit) transmitted from the lowest (LSB) to the most (MSB) significant bit -> the byte needs to be read in reverse!



This way, ASCII characters are transferred for the communication.

Python for laboratory experiments and instrumentation

Py-Serial

[1]:
```
import serial
```

[2]:
```
import serial.tools.list_ports as port_list
ports = list(port_list.comports())
for p in ports: print (p)
```

```
/dev/cu.NokiaAsha501-COM1 - n/a
/dev/cu.Bluetooth-Incoming-Port - n/a
/dev/cu.iPhone-WirelessiAP - n/a
/dev/cu.usbserial-FT3HW45H - US232R - US232R
```

Or a 'COMx' port on Windows, where x is a number attributed by the OS

[3]:
```
serial_address='/dev/cu.usbserial-FT3HW45H'
```

[4]:
```
with serial.Serial(serial_address,baudrate=9600,bytesize=8, parity='N', stopbits=1, timeout=.05) as ser:
#with serial.Serial('/dev/cu.usbserial-PX3IJCKS',baudrate=9600,bytesize=8, parity='N', stopbits=1, timeout=.05) as ser:

    ser.write(b'*IDN?\r')
    l=ser.readline()
    response = l.decode('ascii', errors='replace')
    print(response)
```

```
KEITHLEY INSTRUMENTS INC.,MODEL 2000,4416104,B01  /A02
```

Python for laboratory experiments and instrumentation

```
[4]: with serial.Serial(serial_address,baudrate=9600,bytesize=8, parity='N', stopbits=1, timeout=.05) as ser:
     #with serial.Serial('/dev/cu.usbserial-PX3IJCKS',baudrate=9600,bytesize=8, parity='N', stopbits=1, timeout=.05) as ser:

         ser.write(b'*IDN?\r')
         l=ser.readline()
         response = l.decode('ascii', errors='replace')
         print(response)

KEITHLEY INSTRUMENTS INC.,MODEL 2000,4416104,B01  /A02
```

- The serial address is defined by the OS for the FTDI port
- The baud rate (number of times per second a serial communication signal changes states) is defined on the instrument, can often be chosen, for instance 300, 600, 1200, 2400, 4800, 9600, 19200 ...
- The byte size depends on the instrument protocol and define the number of bits transferred per character
- The parity is a method for detecting errors in transmission: an extra bit (0 or 1) is added to each byte so that the byte is always even (or odd).
- The stop bit, in asynchronous communication, is actually used for timing (or synchronizing) the data characters being transmitted

